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CHAPTER 1

INTRODUCTION

Navigation, the process of accurately estimating position and planning a course of travel, is an important part of vehicle automation. The common backbone of most modern navigation systems consists of an inertial measurement unit (IMU) [1–3]. An IMU is composed of several accelerometers and gyroscopes. These sensors enable the vehicle to estimate attitude and position. If these sensors were perfect, the vehicle could simply employ a process of integrating their measurements to estimate attitude and position called dead reckoning. However, there are inherent problems with these sensors such as misalignments, biases, and scaling factors. These imperfections make navigation by dead reckoning over long periods of time inaccurate and therefore unacceptable.

To remedy this problem, more sensors may be added to the system to provide more information about the navigation states. These sensors, commonly referred to as aiding devices, each provide useful information by way of what is called measurement geometry. Often, each sensor will provide different measurement geometry. These different measurement geometries, if combined intelligently, can help the system by increasing state knowledge and minimizing uncertainty. It has become a common practice to combine this information and calculate a measure of certainty in Kalman Filters.

One of the most common and important aiding systems in modern vehicle automation and navigation is the global positioning system, or GPS. However, GPS can be easily jammed and is often unreliable in environments such as natural and urban canyons. Situations or environments in which GPS is unreliable or denied are commonly referred to as GPS-denied environments. Both military and civilian applications have fueled a growing interest in recent years to find effective GPS-denied navigation systems.

One alternative to GPS-aided systems that has been explored extensively has been vision-aided navigation [4,5]. This method of navigation aiding method used to be imprac-

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tical because of limitations on processing capabilities on board smaller vehicles. However, modern hardware has started to make vision-aided navigation a real possibility and it has been proven to be a plausible mode of navigation when GPS is unavailable [5]. Several ideas and algorithms related to making vision aided navigation work well have been developed in recent years [4].

One promising vision-aided navigation algorithm is the use of angles-only localization. This is where the sensor recognizes landmarks in the environment with known positions and then deducts information useful for navigation from the observed angles to the landmark Fig. 1.1. While promising, the minimum requirements of this exact method of navigation have not been well characterized like GPS pseudorange needs have been. There are several works which explore various landmark placement algorithms which will be discussed in the related work section of this document. None, however, explore both questions proposed by this work.

The rest of this proposal is organized as follows. Chapter 2 explores prior research related to this work. The thesis statement and objectives are defined in chapter 3. The proposed method of research is presented in chapter 4. Finally, an estimated timeline is shown in chapter 5.

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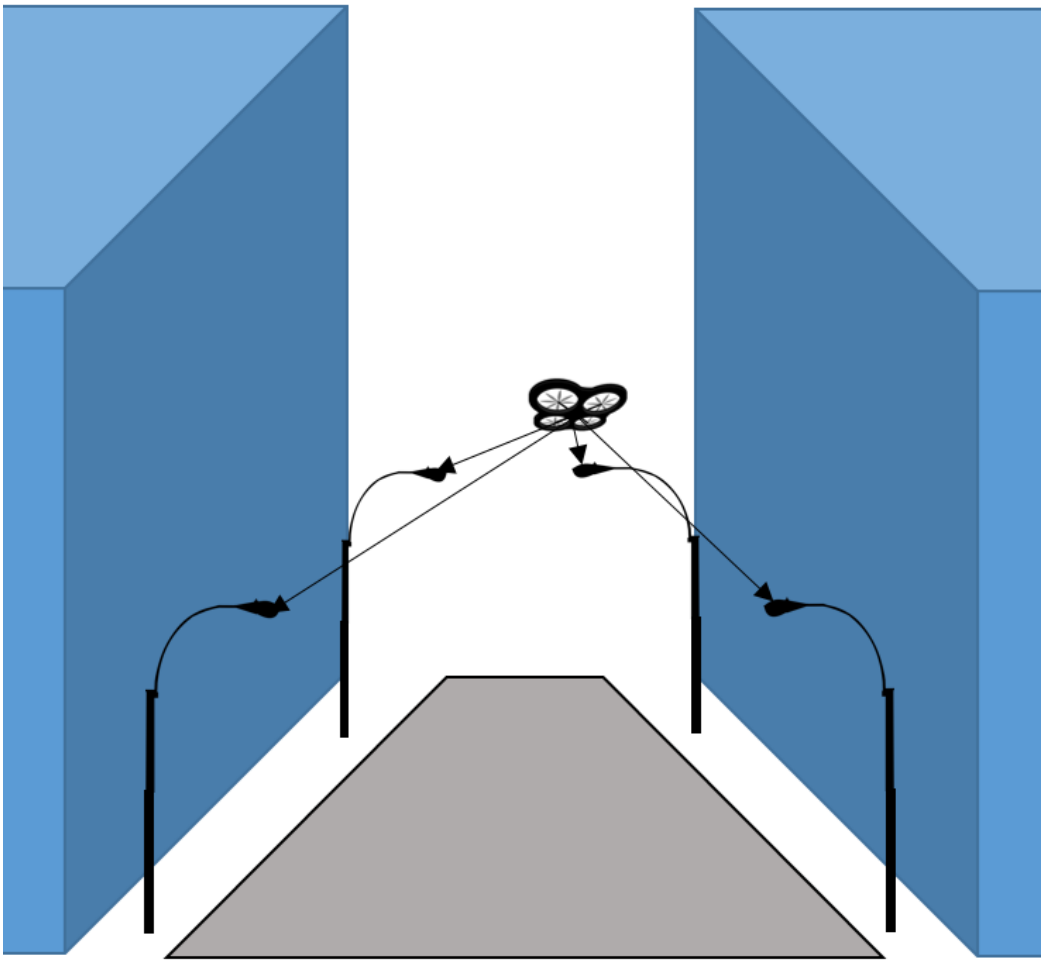


Fig. 1.1: A visual representation of navigation using line-of-sight measurements.

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CHAPTER 2

RELATED WORK

As was mentioned in the introduction section, vision-aided navigation has been shown to be a promising solution for GPS-denied navigation [4,5]. As discussed in the work by Wu et. al., LOS measurement with good measurement geometry gives significant information about vehicle position and attitude as long as landmark positions are known. Due to sensor limitations, the position information may or may not be comparable to position information obtained through GPS measurements. On the other hand, a monocular camera gives better attitude information than a single GPS antenna. Wu et. al. demonstrated this in both software and hardware.

The idea of placing landmarks and sensors for vision-aided navigation is obviously not new. The problem of where to place these landmarks has been examined. Jourdan and Roy studied the placement of static sensors on the walls of buildings to minimize average position error [6]. Sala et. al. chose landmark positions so that a given number were always visible [7]. This requires the solution to include at least one feature within the field of view at all times. This study will examine distributions that do not have this requirement in an effort to find the most cost-effective solutions.

Vitus and Tomlin conducted a similar analysis and presented an optimization method for choosing where to place landmarks to minimize error by estimating the a-priori uncertainties with the experimental covariances of the most likely run of the robot [8]. Beinhofer et. al. attempted to further these works by taking into account the vehicle specifications and its navigating task [9]. Work done by Van den Berg et. al. focuses their efforts mainly on path planning [10].

This work will seek to extend their work by considering both landmark placement and path planning side-by-side. The purpose is still to determine the most cost-effective solution for navigation within the constraints of the LOS measurement method set forth in by Wu

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et. al. [5]. This work will seek to better characterize what the minimum requirements would be to navigate using this method in GPS-denied environments.

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CHAPTER 3
PROPOSED WORK

3.1 Thesis Statement

This work will attempt to provide insight into two questions related to the angles-only flavor of vision-aided navigation as a vehicle travels from point to point. In both questions landmark locations will be known to the vehicle. First, given a fixed travel path and the freedom to place landmarks in any distribution, what is the minimum number of landmarks required to achieve a desired navigation accuracy? Second, given a set distribution of known landmarks, what is the shortest path a vehicle may travel from point to point to achieve a desired navigation accuracy?

Though the derivations and results from this work will be shown using a model of a fixed-wing small unmanned aerial vehicle (UAV), they will apply to all systems that use angles-only localization in vision-aided navigation algorithms.

3.2 Objectives

Over the course of this thesis research, the following 4 objectives will be completed.

1. An extended Kalman filter will be developed to include a line of sight measurement measurement model and this filter will be incorporated into an existing Matlab/Simulink model of an unmanned aerial vehicle.

2. Cost function(s) will be developed to aid in the selection of a landmark distribution and a travel paths.

3. Optimization methods will be applied to the developed cost functions to select a distribution and a travel path to examine.

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4. The proposed solutions selected in objective 3 will be validated using Monte Carlo analysis by ensuring that state estimation remains within the required covariance.

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CHAPTER 4
APPROACH

4.1 Develop and Validate Vehicle and Measurement Model

In order to conduct the study regarding the effect of landmark distribution and travel path on a state uncertainty, a vehicle model must be created to act as a platform for testing. Creating this vehicle model in Matlab/Simulink will be the first task of this work. The first choice to be made will be that of the vehicle to model. Because this work is intended to be broadly applicable, a vehicle with six degrees of freedom (6-DOF) unmanned aerial vehicle (UAV) will be chosen. This is plausible because LOS measurements have already been shown to give usable information in three degrees of position and three degrees of attitude.

Since the focus of this work is not to develop new or innovative vehicle models, it is appropriate to follow previously developed work here. The Matlab/Simulink model for this study will be created by following much of the structure and resources set forth for a fixed-wing small UAV as set forth in Beard and McLain's book, Small Unmanned Aircraft Theory and Practice [3]. The books implementation of a gimbaled camera model may be used to to gather line of sight (LOS) data. Assuming the pointing angles of the gimbal are known, position and attitude information may be obtained from this model. The measurement will be defined as the same measurement used by Wu et. al. [5].

4.2 Develop and Validate Localization Algorithm

Once the vehicle and camera models are implemented and can give a LOS measurement, this information will need to be processed to give localization information. Since this algorithm is not a new idea, it will be rederived for the extended Kalman filter following work done by Wu et. al. [5], taking into account the vehicles camera setup. The resulting derivation will then be implemented into the Matlab/Simulink model and added to the

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extended Kalman filter. Monte Carlo analysis will then be used to validate the algorithm and make sure the filters state estimation is statistically correct [11].

4.3 Develop Cost Function

After the vehicle and measurement models have been implemented and validated, there needs to be a way to determine which distribution and path to examine in this study. In order to do this, two cost functions will be developed: one to help evaluate distributions and one to help evaluate paths. Costs to consider may include items such as cost to place each landmark, fuel required per unit distance of travel path, and state knowledge or uncertainty requirements. More research will be conducted to determine what costs and what type of cost function will best fit the questions addressed by this study.

4.4 Determine Optimal Distribution and Path Using Cost Functions

In order to select a distribution and a path from the infinite set of possibilities to test, optimal options must be identified. Once the cost functions have been developed, the stage will have been set to select optimal solutions. Each cost function will be examined using optimization techniques such as identifying the minima and maxima. This analysis will point to the most cost effective, optimal solutions. These most cost effective solutions will be examined to determine validity.

4.5 Validate Distributions and Paths

The distribution and path will each need to be proven to be a valid solution. Each will be declared valid if it allows the system to operate within a required uncertainty through the duration of travel. There are several options to consider when creating a baseline requirement. One option for selecting required uncertainty limits may be a typical uncertainty while traveling using GPS. Another requirement option may be dependent upon how much space is available to travel in urban canyon situations. One or more of these requirements will be considered when determining validity for all distributions and paths to encourage consistency. Validation will be done through the use of Monte Carlo analysis rather than

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linear covariance analysis because of the highly nonlinear nature of LOS measurements. Results will be recorded, organized, and analyzed.

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CHAPTER 5

TIMELINE

Table 5.1 shows the proposed timeline for research.

Table 5.1: Proposed schedule for research

Milestone	Date
Proposal submission	November 1
Finish implementing UAV model	November 7
Finish implementing LOS measurement for estimation	November 14
Complete development of cost function	November 21
Collect Monte Carlo data for each case	December 15
Results organized and compared	December 31
Thesis final draft complete	February 30

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